Mail – Functional Workflow

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Parent Module: Communication · **Child Module:** 05.2 Mail
Version: 1.0 · **Last Updated:** 31-Oct-2026
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1. Overview

Mail is a private internal mailbox for messages between TaskFlow users. Users can read, search, write, reply to, forward and organise mail, save drafts, and send and download attachments. It does not send to outside email addresses. TaskFlow itself also delivers deadline reminders and task comment alerts into the Inbox as mails from "TaskFlow".

2. Purpose / Business Value

- Staff can write to each other about their work without leaving TaskFlow or using outside email.
- Each mailbox is private to its owner. Read state, stars, flags and trash are personal.
- A badge on the **Mail** menu item shows how many Inbox mails are still unread.
- Reminders and comment alerts can also be kept in the Inbox, alongside personal mail.

3. Who Can Access

Role	Can See	Can Use	Access Level (Own / Team / Organization)
Administrator	✅	✅	Own mailbox; recipient lookup over all active users (Organization)
Manager	✅	✅	Own mailbox; recipient lookup over all active users (Organization)
Employee	✅	✅	Own mailbox; recipient lookup over all active users (Organization)
Custom Role	Configurable (`mail.read`)	Configurable (`mail.compose`, `mail.organize`, `mail.attachments`)	Own, as granted

> No role, including Administrator, can read another person's mail. Every list and every action works on the signed-in user's own mailbox. If a mail belongs to someone else, the screen reports "not found".

4. Screens

#	Screen Name	Menu Path	Purpose	Visible To
1	Mail	Communication Management → Mail (shows an unread badge)	Read, write, reply to, forward and organise internal mail, with attachments	Roles holding `mail.read` (Administrator, Manager, Employee by default)
–	Compose panel (inside Mail)	Mail → **Compose**, **Reply**, **Reply all**, **Forward**, or open a draft	Write a new mail, reply or forward one, or continue a saved draft	Roles holding `mail.compose`

> The original BRD Screen 14 "Email Configuration" (SMTP settings) is marked **SUPERSEDED** in the requirements. It was replaced by this internal Mail screen.

5. Screen & Functionality Details

5.1 Screen: Mail

Purpose: A private internal mailbox for sending and receiving mail between TaskFlow users. It does not send to outside email addresses. **Who can see this screen:** Any role with `mail.read`. By default that is Administrator, Manager and Employee. **Required permission:** `mail.read` to open the screen and read mail. `mail.compose` to write. `mail.organize` to star, flag, trash, restore and delete. `mail.attachments` to upload and download files.

Screen Elements

- **Folder rail:** Inbox, Starred, Important, Sent, Drafts and Trash. Counts appear beside Inbox, Starred and Important (unread) and Drafts (total). A **Compose** button is shown with `mail.compose`.
- **Message list:** a search box ("Search inbox", and so on) that searches subject, text, sender and recipients. Each row shows the sender, or "To: ..." in Sent and Drafts, and "TaskFlow" for system mail. Rows also show the subject (or "(no subject)"), a preview, the date, an important flag, a paperclip when there are attachments, and a star toggle (with `mail.organize`). There are 25 mails per page with Previous / Next.
- **Reading pane:** the subject and an **Open the task** link when the mail is about a task. Below that is the whole conversation (oldest first), with sender, recipients, Cc, date, body text with clickable web links, and attachment download buttons.
- **Reader actions: Reply** and **Reply all** (not shown for system mail or trashed mail), **Forward**, **Star**, **Mark important**, **Mark unread** (Inbox only) and **Move to Trash**. In Trash the actions are **Restore** and **Delete forever**.
- **Compose panel (side drawer)**, titled "New mail", "Reply", "Reply all", "Forward" or "Draft". It has **To** and **Cc** recipient pickers that suggest names as you type and show invalid recipients as "Not a TaskFlow user". It also has **Subject**, **Message**, an **Attach** button, the attached file chips, and the buttons **Discard**, **Save draft** and **Send**.
- **Narrow screens:** one pane at a time, with a folder button to open the folder rail.

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5.2 Functionalities

#	Functionality	Purpose	Who Can See	Who Can Use	Permission Required	Scope
1	Open Mail and browse folders	See Inbox, Starred, Important, Sent, Drafts, Trash and their counts	Administrator, Manager, Employee, Custom (if granted)	Same	`mail.read`	Own
2	Search mail	Find mail in the current folder by subject, text, sender or recipient	Same	Same	`mail.read`	Own
3	Read a mail and its conversation	Open a mail. It is marked read automatically.	Same	Same	`mail.read`	Own
4	Unread badge on Mail menu	See how many Inbox mails are unread	Same	Same	`mail.read`	Own
5	Compose and send a new mail	Write to one or more TaskFlow users (To/Cc)	Roles with `mail.compose`	Same	`mail.compose`	Own (to any active user)
6	Reply / Reply all / Forward	Continue a conversation or pass a mail on	Roles with `mail.compose`	Same	`mail.compose`	Own
7	Save, continue and discard drafts	Finish a mail later	Roles with `mail.compose`	Same	`mail.compose`	Own
8	Recipient suggestions	Find active users by name or email while typing	Roles with `mail.compose`	Same	`mail.compose`	Organization (active user directory)
9	Star / Mark important / Mark unread	Personal flags to find mail later	Roles with `mail.organize`	Same	`mail.organize`	Own
10	Move to Trash / Restore	Remove a mail from folders, or bring it back	Roles with `mail.organize`	Same	`mail.organize`	Own
11	Delete forever	Permanently remove a mail from Trash, or discard a draft	Roles with `mail.organize`	Same	`mail.organize`	Own
12	Attach files	Upload up to 5 files to a mail or draft	Everyone who can compose (the button is not hidden by permission)	Roles with `mail.attachments`	`mail.attachments`	Own uploads
13	Download attachments	Save a file from a mail	Everyone who can read mail (the button is not hidden by permission)	Roles with `mail.attachments`	`mail.attachments`	Own uploads or mails in own mailbox

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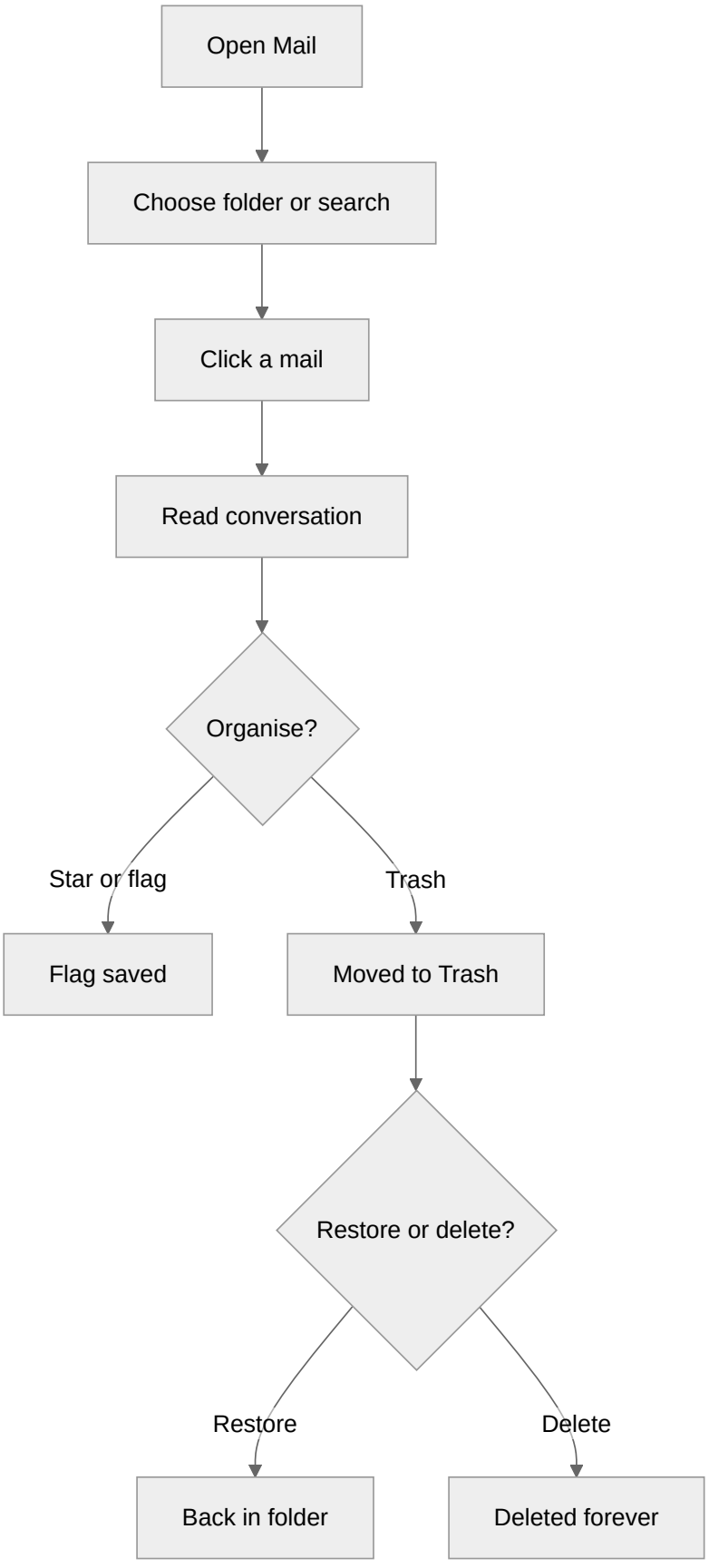
#	Functionality	Purpose	Who Can See	Who Can Use	Permission Required	Scope
14	Receive system mails	Deadline reminders and task comment alerts arrive from "TaskFlow". Reassignment notices are sent as a mail from the person who reassigned.	All mail users	System generated	None. Controlled by the Reminder Settings "Also deliver reminders to Mail" toggle.	Own

5.3 Step-by-Step Workflow – Read and Organise Mail

1. The user opens **Communication Management → Mail**. The Inbox opens by default.
2. The user picks a folder in the rail and can type in the search box.
3. The user clicks a mail. The reading pane shows the conversation and the mail is marked read.
4. The user can **Star**, **Mark important**, **Mark unread** (Inbox), or **Move to Trash**. The system shows *"Moved to Trash"*.
5. In **Trash**, the user can **Restore** (*"Restored"*) or **Delete forever** (*"Deleted forever"*).
6. The folder counts and the Mail menu badge update.

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Flow Diagram



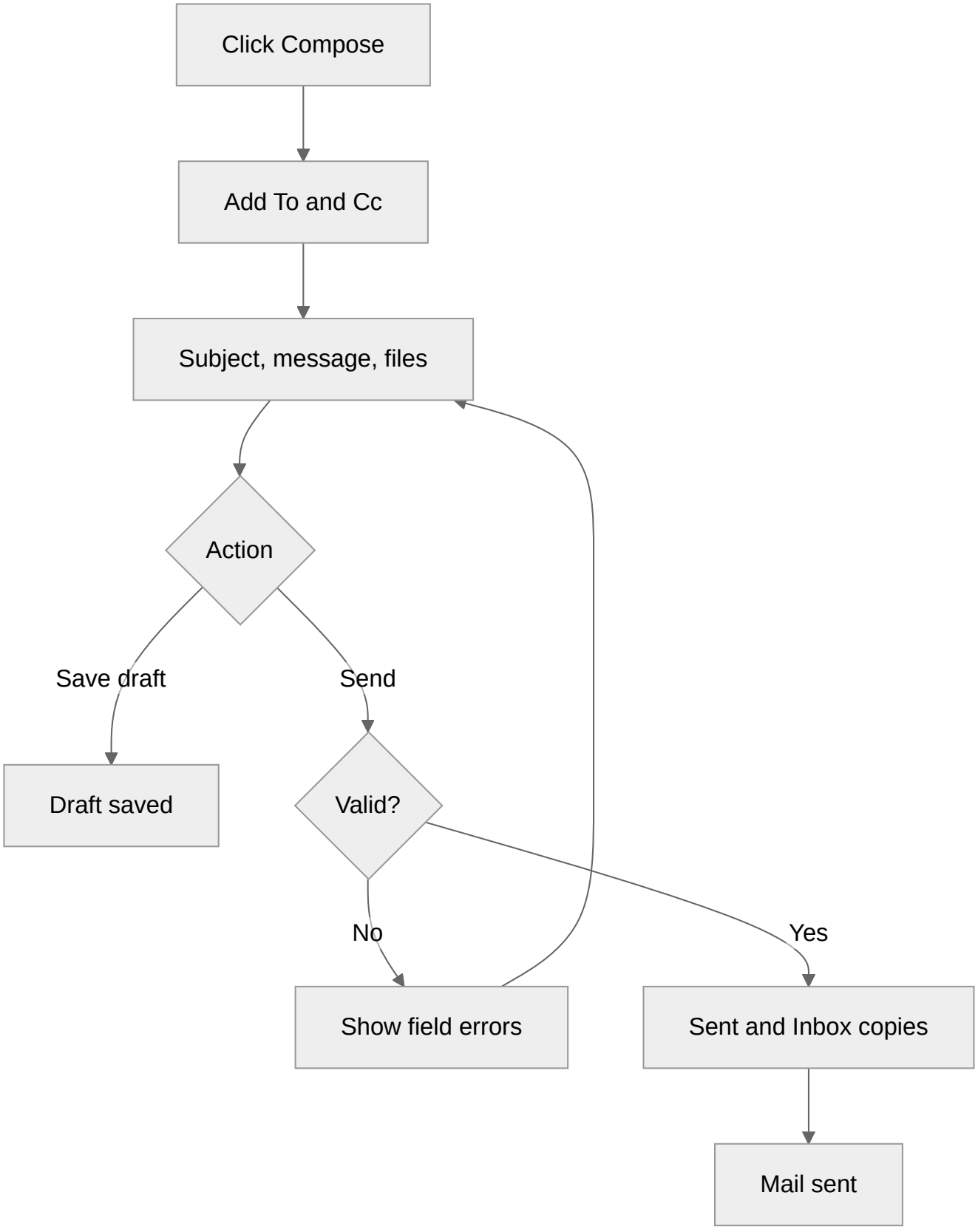
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5.4 Step-by-Step Workflow – Compose and Send a Mail

1. The user clicks **Compose**. The compose panel opens with the title "New mail".
2. The user types names or emails in **To** (and **Cc** after clicking **Cc**). Suggestions appear and each recipient is checked.
3. The user enters a **Subject** and/or **Message**, and can click **Attach** to add files.
4. The user clicks **Send**.
5. The system checks: at least one recipient, every recipient is an active TaskFlow user, a subject or message is present, and attachment limits are respected.
6. If any check fails, the field error is shown under To, Cc, Message or the files. Otherwise the mail is saved in the sender's **Sent** folder and in every recipient's **Inbox**, and the system shows *"Mail sent"*.
7. Instead of sending, the user can click **Save draft** (*"Draft saved"*) or **Discard**. Discard also removes any files uploaded in this panel.

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Flow Diagram



5.5 Step-by-Step Workflow – Reply, Reply All or Forward

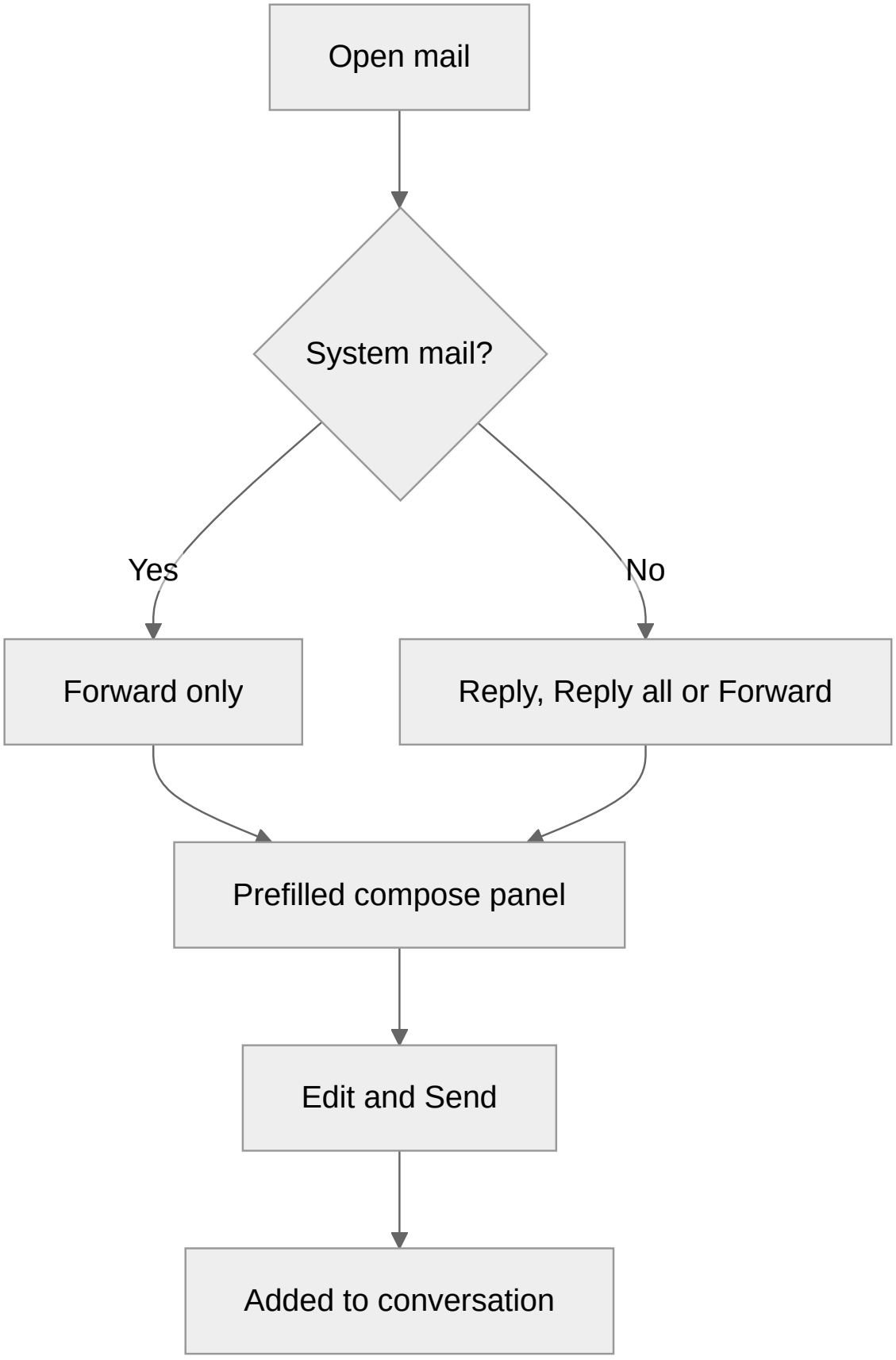
1. The user opens a mail and clicks **Reply**, **Reply all** or **Forward**.
2. The system pre-fills the compose panel:

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    - **Reply:** the original sender (or the original recipients, if you sent it). Subject "Re: ..." and the quoted text "On *date*, *name* wrote:".
    - **Reply all:** the sender plus everyone else on To and Cc, except you.
    - **Forward:** no recipients. Subject "Fwd: ...", a "---------- Forwarded message ----------" header, and the original attachments.
3. The user edits the panel and clicks **Send**. The new mail joins the same conversation.
4. Mails from TaskFlow (reminders, comment alerts) show only **Forward**. Trying to reply gives *"Mails from TaskFlow cannot be replied to. Forward it instead."*

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Flow Diagram



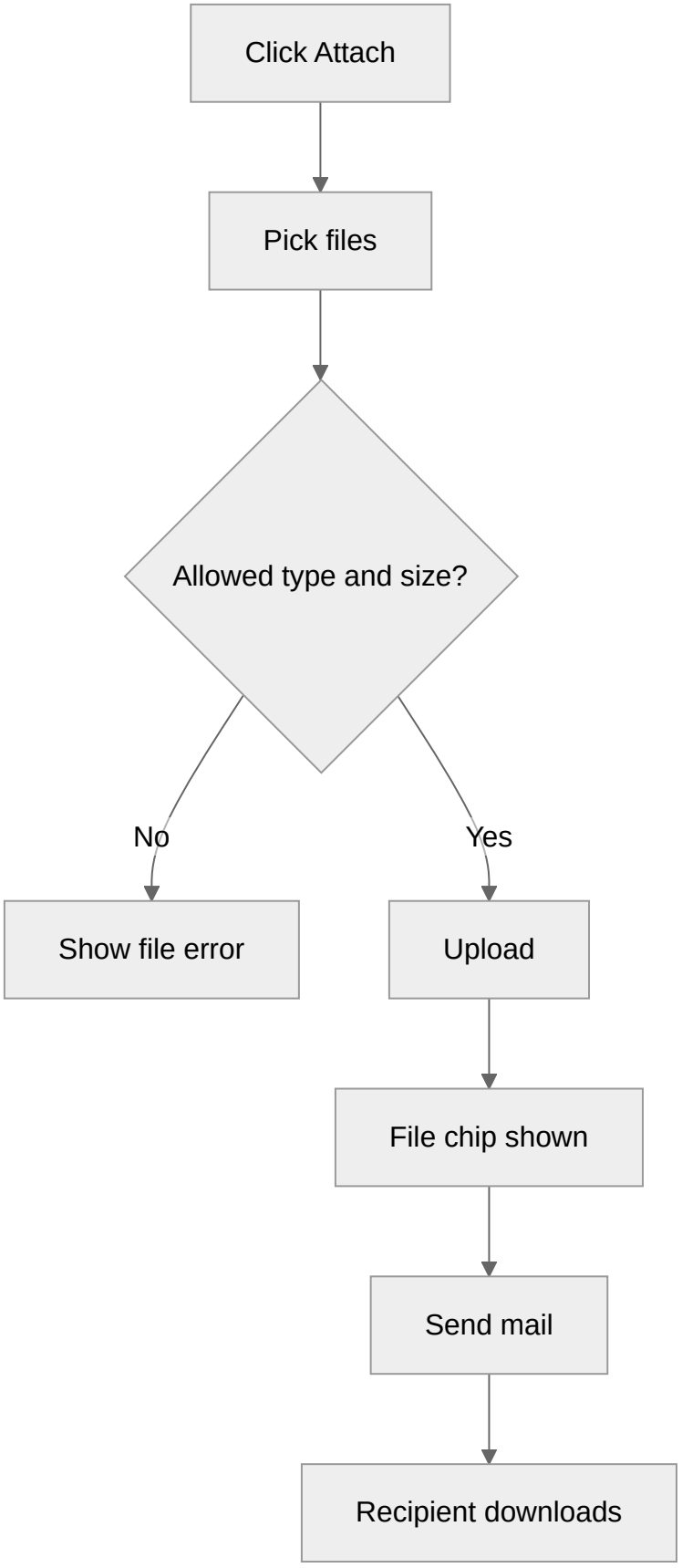
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5.6 Step-by-Step Workflow – Attach and Download Files

1. In the compose panel the user clicks **Attach** and picks one or more files.
2. The browser checks the file type, empty files, the 10 MB per-file limit, the maximum of 5 files and the 25 MB total. The server checks the same limits again.
3. Each accepted file uploads ("Uploading...") and appears as a chip. The user can remove a chip with **x**.
4. When the mail is sent, the files go with it. Forwarded attachments are copied automatically.
5. A reader clicks an attachment in the reading pane to download it. Files are always downloaded, never opened inside the page.

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Flow Diagram



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5.7 Business Rules & Validations

- Mail can only be sent to **active TaskFlow users**. Any unknown address blocks the whole send.
- A mail needs **1 to 50 recipients** in total (To + Cc). A person listed in both To and Cc is kept in To only.
- A mail needs a **subject or a message**. The subject can be up to **200 characters** and the message up to **20,000 characters**.
- A mail can carry at most **5 attachments**, each at most **10 MB**, together at most **25 MB**. Forwarded files count towards these limits.
- Blocked file types: `.exe`, `.dll`, `.bat`, `.cmd`, `.com`, `.msi`, `.ps1`, `.sh`, `.js`, `.vbs`, `.scr`. Empty files are refused.
- A user can have at most **20 files (100 MB)** uploaded but not yet attached to any mail or draft.
- Drafts can be saved with incomplete or invalid recipients. Recipients are checked only when the draft is sent.
- Read state, star, important and trash are **personal**. Changing them affects only your own copy.
- **Delete forever** works only on mail that is already in Trash, or on a draft.
- Mail sent by TaskFlow (reminders, comment alerts) cannot be replied to, only forwarded.
- You can download an attachment only if you uploaded it or the mail is in your mailbox.
- Opening a mail marks it read. Old mail from deactivated or deleted users still shows their name.
- Recipient suggestions show at most 10 active users, matched by name or email.

5.8 Messages Shown to the User

Situation	Message
Mail sent	Mail sent
Draft saved	Draft saved
Draft discarded	Draft discarded
Moved to Trash / Restored / Deleted forever	Moved to Trash · Restored · Deleted forever
No recipients	Add at least one recipient.
Recipient not a user (picker)	Not a TaskFlow user: *addresses*. Mail can only go to people in TaskFlow.
Recipient not a user (on send)	Remove or fix recipients that are not TaskFlow users. / Not an active TaskFlow user: *addresses*.
Recipients still being checked	Checking recipients…
Too many recipients	A mail can have at most 50 recipients. / Add between 1 and 50 recipients.
Empty mail	Write a subject or a message before sending.
Invalid address	'*address*' is not an email address.
Too many attachments	A mail can have at most 5 attachments.
Attachments too large	Attachments on one mail can total at most 25 MB.

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Situation	Message
File too large	*file* is larger than 10 MB. / A file can be at most 10 MB.
Blocked file type	Files of type *.ext* are not allowed.
Empty file	*file* is empty. / The file is empty.
Upload missing	Choose a file to upload.
Too many pending uploads	You have *N* files waiting to be attached. Send or discard a mail before uploading more.
Pending uploads over 100 MB	Your files waiting to be attached already total 100 MB. Send or discard a mail before uploading more.
Attachment not available	One or more attachments are not available.
Reply to system mail	Mails from TaskFlow cannot be replied to. Forward it instead.
Delete forever outside Trash	Move the mail to Trash before deleting it forever.
Empty folder / no search hits	Nothing in *folder*. · No mail matches your search.
Nothing selected	Select a mail to read it.
Load / open / action failures	Could not load your mail. · Could not open this mail. · Could not update the mail. · Could not mark as unread. · Could not download the file. · Could not send the mail. · Could not save the draft. · Could not prepare the reply. · Could not open the draft. · Could not discard the draft. · Could not upload *file*.
No permission (open screen)	The user is returned to their dashboard; no message is shown.
No permission (action)	The action fails with the related "Could not ..." message (the server returns 403 Forbidden).

6. Permission Summary

Permission Key	Description	Administrator	Manager	Employee	Custom Role
`mail.read`	Read mail: open the inbox, folders and messages; Mail menu badge	Granted	Granted	Granted	Configurable
`mail.compose`	Write and send mail: compose, save drafts, reply and forward, recipient lookup	Granted	Granted	Granted	Configurable
`mail.organize`	Organize mail: star, mark important, move to trash, restore and delete	Granted	Granted	Granted	Configurable
`mail.attachments`	Use mail attachments: upload, download and remove attachments	Granted	Granted	Granted	Configurable

> All four keys are granted by default to Administrator, Manager and Employee, and to every role that existed when the permissions were first seeded. A new custom role starts with none of

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> them, and an administrator can grant them (or copy them from another role) in Role Configuration.

7. Related Modules

- **Notifications (05.1, sibling child module)** – the reminder engine delivers deadline, due-today and overdue reminders into the Inbox as mails from "TaskFlow" when "Also deliver reminders to Mail" is on.
- **Monitoring (Task Management, Comments, Reassignment)** – comments create mails from "TaskFlow", and reassignments send a mail from the person who reassigned. A mail about a task shows an **Open the task** link.
- **Settings (Reminder Settings, Application Settings)** – the Reminder Settings "Also deliver reminders to Mail" toggle controls system mails. The application timezone is used for the dates quoted in replies and forwards.
- **TaskFlow Assistant (TFA)** – the assistant can find users by name for a mail and draft a mail that the user reviews and confirms before it is sent.
- **Settings → Role Configuration** – grants or removes the four `mail.*` keys for custom roles.

8. Expected Outcome

Users can exchange private mail with colleagues inside TaskFlow, including replies, forwards, drafts and attachments, keep their mailbox organised with personal flags and trash, and see unread mail at a glance, without anyone else being able to read their mailbox.

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